# Mock test (examination week): solutions

**Self-marking.** Mark your own answers part by part with the marks in square brackets. Parts with the lifebuoy 🛟 add up to 30 marks out of 50: with all of them you are at the pass mark (60%). Write your total, and the total of the lifebuoy parts, at the top of your test.

| Exercise | Marks | of which 🛟 |
|---|---|---|
| 1 Multiple choice | 10 | 10 |
| 2 True or false | 4 | 4 |
| 3 A delivery robot | 8 | 4 |
| 4 A stone thrown upwards | 9 | 4 |
| 5 Energy resources | 5 | 5 |
| 6 Pushing a crate | 7 | 2 |
| 7 An electric hoist | 7 | 1 |
| **Total** | **50** | **30** |

## Exercise 1 — Multiple choice (10 marks, all 🛟)

1. **C.** $v = s/t = 90/1.25 = 72\,\text{km/h}$, because 1 h 15 min = 1.25 h. A ignores the 15 minutes, B multiplies $90 \times 1.25$, D is 90/75: km per minute, not per hour.
2. **A.** The gradient of a distance–time graph is the speed, so a steeper line means faster. B and D confuse the position and the duration with the gradient; C needs a curve.
3. **D.** $E_k = \frac{1}{2}mv^2$ is proportional to $v^2$: the speed is multiplied by 3, so the kinetic energy by $3^2 = 9$.
4. **B.** $\Delta E_p = mg\Delta h = 4.0 \times 10 \times 2.5 = 100\,\text{J}$. A is $m \times h$, C is the weight $mg = 40\,\text{N}$ (a force, not an energy), D has an extra factor of 4.
5. **C.** At the top the stone stops: $E_k = 0$ and the gravitational potential energy is at its maximum. The total energy never changes and is not zero (B); the two stores are equal only at one particular height (D).
6. **B.** $W = Fd = 40 \times 3.0 = 120\,\text{J}$: the time does not enter. A is 40/5.0, C is $40 \times 5.0$, D is $40 \times 3.0 \times 5.0$.
7. **A.** $W = mg\Delta h = 20 \times 10 \times 3.0 = 600\,\text{J}$, then $P = W/t = 600/4.0 = 150\,\text{W}$. B forgets to divide by the time, C is only the weight in newtons, D multiplies by the time.
8. **D.** The total input is $90 + 30 = 120\,\text{kJ}$, so the efficiency is $90/120 = 75\%$. A is the wasted fraction, B is 30/90, C is 90/30.
9. **C.** Efficiencies in a chain multiply: $0.80 \times 0.50 = 0.40 = 40\%$. B is the mean, A the difference, D the sum.
10. **B.** Geothermal energy comes from the hot interior of the Earth. Wind, biofuel and hydroelectric power all depend on energy from the Sun.

## Exercise 2 — True or false (4 marks, all 🛟)

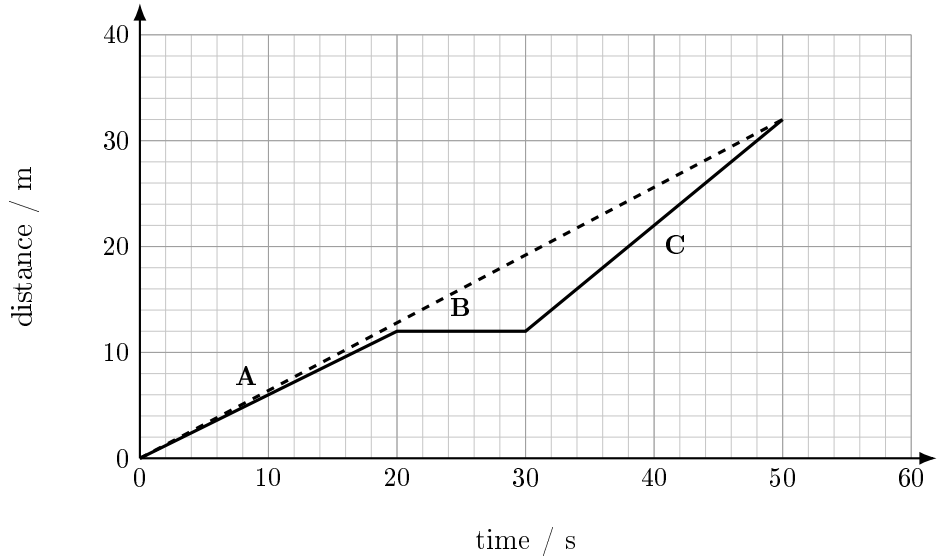
(a) **F**: a horizontal line means the object is stationary. (b) **T**: $1\,\text{W} = 1\,\text{J/s}$. (c) **T**: at the same speed $E_k$ is proportional to the mass. (d) **F**: a 60% efficient machine wastes 40%.

## Exercise 3 — A delivery robot (8 marks)

**Solution** — **(a)** ⚽ Stationary: **B**. Greatest speed: **C** (the steepest line). **[1 + 1]**
**(b)** ⚽ The gradient is the speed: change in distance divided by change in time. **[1]**
**(c)** $v = \dfrac{s}{t} = \dfrac{12}{20} = 0.60\,\text{m/s}$ **[1 for the equation with the values, 1 for the answer]**
**(d)** Total distance 32 m, total time 50 s, including the stop:

$$v_{\text{avg}} = \frac{32}{50} = 0.64\,\text{m/s}$$

**[1 for the two values, 1 for the answer]**
**(e)** ⚽ A straight line from $(0, 0)$ to $(50\,\text{s}, 32\,\text{m})$, as in the figure (dashed). **[1]**




**Check and common mistakes** — (c) $12 \times 20 = 240$ (multiplied instead of divided) and $12/50 = 0.24$ (total time used for the first section). (d) The mean of the three section speeds, $(0.60 + 0 + 1.0)/3 = 0.53\,\text{m/s}$, and $32/40 = 0.80\,\text{m/s}$ (the stop left out): the right value, 0.64 m/s, lies between the speeds of A and C and is smaller than the speed in C.

## Exercise 4 — A stone thrown upwards (9 marks)

**Solution** — **(a)** ⚽ The kinetic energy store decreases and the gravitational potential energy store increases. **[1]**

**(b)** ⚽ The kinetic energy at launch becomes gravitational potential energy at the top:

$$\tfrac{1}{2}mu^2 = mgh \;\Rightarrow\; h = \frac{u^2}{2g}$$

**[1 for the energy equality, 1 for $h$]**

**(c)** $h = \dfrac{u^2}{2g} = \dfrac{12^2}{2\times 10} = 7.2\,\text{m}$ **[1 for the substitution, 1 for the answer]**

**(d)** At 3.0 m the stone has gained $mg\Delta h$ of gravitational potential energy, taken from its kinetic energy:

$$\tfrac{1}{2}mv^2 = \tfrac{1}{2}mu^2 - mg\Delta h \;\Rightarrow\; v^2 = u^2 - 2g\Delta h = 144 - 2\times 10\times 3.0 = 84$$

$$v = \sqrt{84} = 9.2\,\text{m/s}$$

**[1 for the energy equation, 1 for $v^2$, 1 for the answer]**

*Same result by another route:* the stone is $7.2 - 3.0 = 4.2$ m below its highest point, where $v = 0$, so $v^2 = 2g \times 4.2 = 84$.

**(e)** ⚽ The same. The mass cancels: $h = u^2/(2g)$ does not contain $m$. **[1]**

**Check and common mistakes** — (c) Without squaring the speed, $12/20 = 0.60$ m; with $v^2/g$ (the 2 forgotten), 14.4 m. (d) Equating $mg \times 3.0$ with $\frac{1}{2}mv^2$ gives $\sqrt{60} = 7.7$ m/s, which is the speed gained by falling 3.0 m, not the speed left at that height. Check: 9.2 m/s is less than the launch speed of 12 m/s and the stone is still below the top, so it is still moving.

## Exercise 5 — Energy resources (5 marks, all ⚽)

**(a)** 1 mark for each row (both cells correct):

| **Resource** | **R or N** | **Energy store** |
|---|---|---|
| hydroelectric | R | gravitational potential |
| natural gas | N | chemical |
| geothermal | R | thermal (hot rocks inside the Earth) |
| nuclear fission | N | nuclear |

**(b)** ⚽ Any one: no carbon dioxide released while generating electricity; renewable, so the wind does not run out; no fuel to buy. **[1]**

## Exercise 6 — Pushing a crate (7 marks)

**Solution** — **(a)** ⚽ $W = Fd$ **[1]**
**(b)** $W = Fd = 80 \times 6.0 = 480\,\text{J}$ **[1 for the substitution, 1 for the answer with the unit]**
**(c)** ⚽ The thermal energy store of the floor and the crate: friction heats them. The speed is constant, so there is no gain in kinetic energy. **[1]**
**(d)** Without friction all the work goes into kinetic energy:

$$W = \Delta E_k = \tfrac{1}{2}mv^2 \;\Rightarrow\; 480 = \tfrac{1}{2} \times 25 \times v^2 \;\Rightarrow\; v^2 = \frac{2 \times 480}{25} = 38.4$$

$$v = \sqrt{38.4} = 6.2\,\text{m/s}$$

**[1 for $W = \Delta E_k$, 1 for $v^2$, 1 for the answer]**
**Check and common mistakes** — (b) The weight used instead of the pushing force: $250 \times 6.0 = 1500\,\text{J}$. (d) The factor 2 forgotten, $\sqrt{480/25} = 4.4\,\text{m/s}$; the square root forgotten, $38\,\text{m/s}$ (about $140\,\text{km/h}$ for a pushed crate: not reasonable). Check: $6.2\,\text{m/s}$ is about a running speed, and $480\,\text{J}$ of energy for a $25\,\text{kg}$ crate gives a speed of that order.

## Exercise 7 — An electric hoist (7 marks)

**Solution** — **(a)** $\Delta E_p = mg\Delta h = 120 \times 10 \times 4.0 = 4800\,\text{J}$ **[1 for the substitution, 1 for the answer]**
**(b)** $P = \dfrac{W}{t} = \dfrac{4800}{8.0} = 600\,\text{W}$ **[1 for the substitution, 1 for the answer]**
**(c)** $\text{efficiency} = \dfrac{\text{useful power out}}{\text{total power in}} \times 100\% = \dfrac{600}{800} \times 100\% = 75\%$ **[1 for the substitution, 1 for the answer]**
**(d)** ⚽ It is transferred to the thermal energy store of the motor and the surroundings (and some as sound). **[1]**
**Check and common mistakes** — (a) The $g$ forgotten: $120 \times 4.0 = 480\,\text{J}$. (b) The time forgotten: $4800\,\text{W}$ (a power far above the $800\,\text{W}$ supplied). (c) The ratio upside down: $800/600 = 133\%$; an efficiency above 100% is impossible, so the ratio must be useful over total.